

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 5: System Software

Subtopic 5.1: Operating Systems

Concept 5.1.2: OS Management Tasks

Total Questions: 12

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q5.a.i)

5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(a) (i) State **two** tasks that are performed by hardware management.

1

.....

2

.....

[2]

Question 2 | Source: 9618_w25_qp_11 (Q5.a.ii)

5 The Operating System (OS) is responsible for hardware and security management in a computer system.

(ii) State **two** tasks that are performed by security management.

1

.....

2

.....

[2]

Question 4 | Source: 9618_w24_qp_11 (Q4.a.i)

4 A student uses a laptop to write a program that is saved as a text file.

(a) The laptop has utility software and an Operating System (OS).

(i) Describe the file management tasks carried out by an OS.

.....

.....

.....

.....

[2]

Question 5 | Source: 9618_w24_qp_12 (Q3.c)

3 A student has a computer.

- (c) The computer has an Operating System (OS). One of the key management tasks of the OS is process management.

Describe the process management tasks performed by an OS.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 6 | Source: 9618_s24_qp_12 (Q6)

- 6 A computer has an Operating System (OS).

Memory management and process management are two OS tasks.

Explain how memory management **and** process management support multi-tasking.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 7 | Source: 9618_s23_qp_11 (Q3.a)

- 3 A computer has an Operating System (OS).

- (a) Describe how the Operating System manages the peripheral hardware devices of the computer.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Question 8 | Source: 9618_s23_qp_11 (Q3.b)

- 3 A computer has an Operating System (OS).

- (b) Hardware management is one key management task carried out by the Operating System.

Identify **two other** key management tasks carried out by the Operating System.

1

2

[2]

Question 9 | Source: 9618_s23_qp_12 (Q5.d.i)

- 5 A student has purchased a new laptop.

- (d) The laptop has systems software.

(i) Describe how the Operating System (OS) manages processes in the computer.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

Question 10 | Source: 9618_w22_qp_11 (Q7.b)

(b) Memory management is one of the tasks performed by an Operating System (OS).

Describe the ways in which memory management organises and allocates Random Access Memory (RAM).

.....

.....

.....

..... [2]

Question 11 | Source: 9618_w22_qp_13 (Q3)

3 Draw **one** line from each Operating System (OS) management task to its most appropriate description.

OS Management task	Description
hardware management	dynamically allocates memory to processes
security management	marks unallocated file storage for availability
memory management	installs programs for devices connected to external ports
process management	validates user and process authenticity
	allows processes to transfer data to and from each other

[4]

Question 12 | Source: 9618_w21_qp_12 (Q7.b)

7 A computer has system software.

(b) Describe the file management tasks that an Operating System performs.

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Question 13 | Source: 9618_s21_qp_11 (Q2.b)

2 Bingwen’s computer comes with an Operating System and utility software.

(b) Identify **four** key management tasks that the Operating System will perform.

1

2

3

4

[4]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q5.a.i)

5(a)(i)	1 mark per bullet point, max 2 marks • Installs <u>driver</u> software for devices connected to external ports • Manages communication between devices • Manages <u>hardware</u> interrupts	2
---------	---	----------

Mark Scheme for Question 2 | Source: 9618_w25_ms_11 (Q5.a.ii)

5(a)(ii)	1 mark per bullet point, max 2 marks • Prevents unauthorised access • ... by providing authentication // by validating users and processes • Implements access rights and permissions • Makes provision for recovery of lost data • Carries out Operating System security updates as available • Carries out auditing and keeps logs of activity	2
----------	---	----------

Mark Scheme for Question 3 | Source: 9618_w25_ms_13 (Q8.d)

8(d)	1 mark for each bullet point, max 3 marks • File management • Security management • Hardware / peripheral management // input / output management • Process management	3
------	---	----------

Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q4.a.i)

4(a)(i)	1 mark for each bullet point (max 2) e.g. • Secondary storage space is divided into file allocation units • Space on secondary storage is allocated to particular files • Maintains/creates directory structures • Specifies the logical method of file storage e.g. FAT or NTFS • Provides file naming conventions • Controls user access // implements access rights //implements password protection // Makes file sharing possible • Controls access to the file by other software • Specifies tasks that can be performed on a file e.g. open, close, delete, copy, create, move, etc • Allows searching for a file	2
---------	---	----------

Mark Scheme for Question 5 | Source: 9618_w24_ms_12 (Q3.c)

3(c)	1 mark for each bullet point (max 4) e.g. • It manages the scheduling of processes // Decides which process is to be run next • Allows multi-tasking/multi-processing • Ensures fair access • Handles interrupts • Manages / allocates which resources the processes require • Facilitates the sharing and exchange of data between processes • Prevents interference between processes // conflict resolution	4
------	--	----------

Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q6)

6	1 mark each to max 4 Max 3 marks for each management task: <i>Memory management: Max 3 marks</i> • Stores data from all currently running programs concurrently in RAM • Stops the data from overwriting each other in RAM/primary storage • Decides which processes should be in main memory • Makes efficient use of memory <i>Process management: Max 3 marks</i> • Allows one process to be paused whilst another process can be actioned • Decides which process is to be run next • Switches between processes to allow them to share the use of the processor • Identification/description of scheduling	4
---	--	----------

Mark Scheme for Question 7 | Source: 9618_s23_ms_11 (Q3.a)

3(a)	1 mark each to max 4 Examples: • Installs device drivers • ... to allow communication between peripherals and computer • Sends data and receives data to and from peripherals • ... such as to an output device and from an input device/by example • Handles buffers for transfer of data • ... to ensure smooth transfer between devices that transmit and receive at different speeds • Manages interrupts / signals from the device	4
------	--	----------

Mark Scheme for Question 8 | Source: 9618_s23_ms_11 (Q3.b)

3(b)	1 mark each to max 2 • Memory management • File management • Security management • Process management • Error checking and recovery	2
------	---	----------

Mark Scheme for Question 9 | Source: 9618_s23_ms_12 (Q5.d.i)

5(d)(i)	1 mark each to max 5 • Manages the scheduling of processes // decides which order to run processes • Manages which resources the processes require • ... such as allocating memory • Enables processes to share data • Prevents interference between processes // resolution of conflicts • Handles the process queue • It allows multi-tasking / multi-processing • ... by ensuring fair access, handling priorities and handling interrupts	5
---------	--	----------

Mark Scheme for Question 10 | Source: 9618_w22_ms_11 (Q7.b)

7(b)	1 mark for each bullet point (max 2): • RAM is assigned into blocks • dynamic allocation of RAM to programs / processes • reclaims unused blocks of RAM • prevents two programs / processes occupying the same area of RAM at the same time • moves data from secondary storage when needed // manages paging, segmentation and virtual memory	2
------	--	----------

Mark Scheme for Question 11 | Source: 9618_w22_ms_13 (Q3)

3

1 mark for each correct line:

4

OS Management task

Description

hardware
management

dynamically allocates
memory to processes

security management

marks unallocated file
storage
for availability

memory management

installs programs for
devices connected to
external ports

process management

validates user and
process authenticity

allows processes to transfer
data to and from each other

Mark Scheme for Question 12 | Source: 9618_w21_ms_12 (Q7.b)

7(b)	1 mark for each bullet point to max 4 - Storage space is divided into file allocation units - Space is allocated to particular files - Maintains / creates directory structures - Specifies the logical method of file storage (e.g. FAT or NTFS) - Provides file naming conventions - Controls access // implements access rights // implements password protection // Makes file sharing possible - Specifies tasks that can be performed on a file (e.g. open, close, delete, copy, create, move etc.)	4
------	--	---

Mark Scheme for Question 13 | Source: 9618_s21_ms_11 (Q2.b)

2(b)	1 mark per bullet point to max 4 • memory management • file management • security management • hardware / device / peripheral / resources management • input/output management • process management • error checking and recovery • provision of a platform for software • provision of a user interface	4
------	--	----------